

RAJALAKSHMI ENGINEERING COLLEGE

Department of Computer Science and Engineering

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MINI PROJECT REPORT

On

**Flood Damage Area Detection
and Mapping
using Computer Vision and Image Processing**

Individual Report — Algorithm D: GrabCut + HSV Water Range

Submitted by

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Date of Submission: [MAY 2026]

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1. INTRODUCTION

1.1 Background and Motivation

Floods rank among the most frequent and destructive natural disasters worldwide, causing extensive damage to human life, property, infrastructure, and the environment. In India, particularly in regions like Tamil Nadu, Kerala, Assam, and Bihar, recurrent flooding during monsoons leads to significant socio-economic losses every year. According to global disaster databases, floods account for approximately 40% of all natural disasters and affect millions of people annually.

Traditional manual methods of flood assessment are time-consuming and inefficient. With the advancement of Computer Vision and image processing techniques, automated flood detection using RGB imagery has become feasible. Classical image processing methods, such as HSV color segmentation combined with GrabCut refinement, offer computationally lightweight and interpretable alternatives to deep learning approaches. This project focuses on detecting flood-affected regions using these techniques to assist in faster and more efficient disaster management.

1.2 Problem Statement

Given RGB flood images, accurately segment flooded (water-covered) regions from non-flooded areas, handling challenges such as varying lighting, reflections, shadows, and muddy water.

1.3 Objectives

The primary objectives of this mini-project are as follows:

- Implement a combined HSV color masking + GrabCut segmentation pipeline using OpenCV.
- Apply a 10-step preprocessing pipeline (resize, crop, grayscale, blur, threshold, edges, normalize, histogram equalization, morphology, denoise) to enhance image quality.
- Detect muddy/brown water and dark flood regions using multi-range HSV masking.
- Refine detected flood regions using GrabCut algorithm for precise boundary delineation.
- Classify flood severity into Low, Medium, and High levels based on flood area percentage.
- Save and display results including the flood mask and annotated output image.

1.4 Scope of the Report

This report focuses on the HSV + GrabCut flood detection algorithm applied on locally stored RGB flood images.

The scope is limited to:

- Offline processing of static RGB images (street-level and aerial views)
- Use of traditional image processing techniques only (no neural networks or deep learning)
- Performance evaluation on a representative subset of the dataset using Dice and IoU metrics
- Detailed documentation of preprocessing, algorithmic stages, results, and visual outputs

Deep learning methods are out of scope:

- Real-time video processing
- Deep learning-based segmentation methods (U-Net, DeepLab, etc.)
- Multi-spectral or SAR image analysis
- Deployment on embedded hardware or mobile applications

2. LITERATURE REVIEW

HSV-based color segmentation and GrabCut refinement represent well-established classical techniques in computer vision for image partitioning, particularly valuable in scenarios requiring interpretable results without training data. In flood detection, these methods serve as efficient, lightweight alternatives to deep learning models, especially for RGB images where computational resources may be limited during disaster response.

2.1 Methods

The HSV (Hue, Saturation, Value) color space has been widely used for water body detection due to its separation of chromatic and luminance information. Saturation and hue channels are particularly effective in detecting brown or muddy flood water, as well as dark stagnant water regions. The GrabCut algorithm, introduced by Rother et al. [1], is an interactive foreground-background segmentation method that uses Gaussian Mixture Models (GMM) and graph cuts to iteratively refine a given binary mask, producing significantly improved segmentation boundaries with minimal user input.

2.2 Architectures

Morphological operations such as opening, closing, and connected component analysis have been used extensively in conjunction with color-based segmentation to remove noise, fill gaps, and filter out small spurious regions. Such operations, described comprehensively by Gonzalez and Woods [2], are fundamental to improving the quality of binary masks generated by thresholding approaches. The removal of the upper image region (top 1/3) as a heuristic for excluding buildings and sky follows region-of-interest masking approaches commonly used in traffic and aerial scene analysis.

2.3 Existing Method Studies

Flood detection in satellite and aerial imagery has been approached using both classical and deep learning methods. Huang et al. [3] proposed WaterDetectionNet, a CNN-based method for SAR flood mapping. Saleh et al. [4] introduced a transformer-based approach for multi-temporal SAR change analysis. However, for ground-level RGB images with muddy water, classical color-space methods remain relevant and practical. The Kaggle Flood Area Segmentation dataset [5] provides RGB images paired with binary ground truth masks, enabling performance evaluation of classical methods without a training phase.

2.4 Gap Identified

While deep learning methods achieve superior segmentation scores ($\text{Dice} > 0.90$) on SAR or multispectral datasets, they require extensive labeled data, GPU resources, and lack interpretability. Classical HSV + GrabCut pipelines, despite their advantages in speed and explainability, are underexplored for street-level RGB flood images with challenges like reflections, shadows, and muddy water. This project addresses that gap by implementing an enhanced HSV masking and GrabCut refinement pipeline with morphological post-processing, establishing a competitive classical baseline for flood detection.

3. DATASET DESCRIPTION

3.1 Kaggle Dataset

The dataset used in this project consists of locally stored RGB flood images organized in a folder named dataset/flood/. These images represent diverse flood scenarios including street-level views, aerial perspectives, and ground-level scenes featuring muddy water, vegetation, buildings, and varying lighting conditions. The images are processed directly using the OpenCV imread() function.

Each image in the dataset is processed individually through the full pipeline. The results, including the flood mask and annotated output, are saved to an output/ directory organized by filename.

3.2 Data Split

Since this project is based on classical image processing and does not involve model training, a traditional train-validation-test split is not required. All images in the dataset/flood/ folder are processed sequentially. For each image:

1. The image is loaded and validated (skipped if unreadable).
2. Preprocessing is applied and intermediate outputs are saved.
3. Flood detection is performed and the flood percentage and level are computed.

Results are saved and displayed interactively with keyboard control.

Table 1: Dataset Statistics

Dataset	Total Image	Total Masks	Purpose
Kaggle Flood Area Segmentation	290	290	Flooded region segmentation
Used in this project (evaluation subset)	100	100	Quantitative metrics & ablation study

Table 2: Evaluation Approach

Approach	Images Used	Percentage of Dataset	Purpose
Full dataset (visualization & qualitative)	290	100%	Inspection of preprocessing & segmentation
Quantitative evaluation (Dice, IoU)	100	≈34%	Representative performance reporting
Ablation study (preprocessing steps)	50–100	Variable	Analysis of individual stage contributions

4. PREPROCESSING METHODOLOGY

A comprehensive 10-step preprocessing pipeline was designed and applied uniformly to all input RGB flood images before feeding them into the flood detection algorithm. The pipeline is implemented in preprocessing.py and produces intermediate outputs saved at each stage for visual inspection. These steps are critical for enhancing image quality, reducing noise, and preparing the image for effective HSV-based flood detection.

4.1 Grayscale Conversion

The original RGB image is converted to a single-channel grayscale image using the standard luminance formula:

$$I = 0.299R + 0.587G + 0.114B$$

This step reduces dimensionality and focuses on intensity variations, which are often sufficient to separate water (typically darker) from non-water regions in flood scenes.

4.2 Gaussian Denoising

A Gaussian blur filter is applied to suppress high-frequency noise and small artifacts that could lead to spurious local minima during watershed flooding.

- Kernel size: 5×5
- Standard deviation (σ): 0 (automatically computed)

This smoothing operation helps create cleaner intensity gradients while preserving the overall structure of larger flooded areas.

4.3 CLAHE Contrast Enhancement

Contrast Limited Adaptive Histogram Equalization (CLAHE) is applied to improve local contrast, particularly in regions with low dynamic range (e.g., overcast skies, shadowed areas, or muddy water).

- Clip limit: 2.0
- Tile grid size: 8×8

Unlike global histogram equalization, CLAHE operates on small tiles and limits amplification of noise, making subtle differences between flooded and dry surfaces more pronounced — a crucial improvement for subsequent thresholding.

4.4 Binary Segmentation

Otsu's method is used to automatically determine an optimal global threshold that minimizes intra-class variance and maximizes inter-class variance between foreground (possible flooded regions) and background.

- Threshold type: Binary (THRESH_BINARY + THRESH_OTSU)
- Input: CLAHE-enhanced grayscale image
- Output: Initial binary mask (255 = candidate flooded pixels, 0 = background)

This step produces a coarse separation that serves as the foundation for marker extraction.

4.5 Morphological Noise Removal (Opening)

Morphological opening is performed to eliminate small white noise blobs, thin protrusions, and disconnected regions that do not represent meaningful flooded areas.

- Structuring element: 3×3 square kernel
- Iterations: 2

Opening = Erosion followed by Dilation This operation smooths the binary mask and removes isolated false positives caused by reflections or small bright objects, resulting in more compact and reliable candidate regions for marker generation.

4.6 Sure Background Identification

The cleaned binary mask is dilated using a 3×3 kernel (3 iterations) to expand the background regions. This creates a sure background area — pixels that are very likely to be non-flooded — by conservatively growing known background regions inward from the borders and away from potential objects.

4.7 Sure Foreground Extraction (Distance Transform)

A distance transform (`cv2.DIST_L2`, mask size 5) is computed on the cleaned binary mask to measure the distance of each foreground pixel from the nearest background pixel.

The resulting distance map is thresholded at $0.4 \times$ its maximum value to retain only the central, most confident parts of objects. These pixels form the sure foreground (confident flooded region centers). The threshold factor (0.4) ensures only strong, interior regions are marked as definite flood, avoiding uncertain boundary pixels.

4.8 Median Blur

A median blur with a kernel size of 5 is applied to the histogram-equalized image to reduce salt-and-pepper noise while preserving edges. The denoised image is then converted back to BGR format and returned as the final preprocessed output.

5. ALGORITHM: GrabCut + HSV WATER RANGE

5.1 Algorithm Overview

The flood detection algorithm implemented in this project combines HSV color space masking with GrabCut refinement, implemented in `utils.py`. The algorithm operates in a multi-stage pipeline that progressively refines the flood region detection:

- **HSV Color Masking:** Targets muddy water (brown/orange hues, Hue 5-30) and dark water regions (low brightness) using two complementary HSV ranges.
- **Vegetation Removal:** Suppresses green vegetation (trees, grass) using a dedicated green HSV mask to reduce false positives.
- **Morphological Cleaning:** Eliminates noise and fills gaps using morphological opening and closing with a 7×7 kernel.
- **Connected Component Filtering:** Removes small isolated regions below 800 pixels to eliminate spurious detections.
- **Regional Masking:** Suppresses the top 1/3 of the image to exclude buildings and sky from detection.
- **GrabCut Refinement:** Uses the HSV mask as initialization to run the GrabCut algorithm (5 iterations) for precise boundary delineation.
- **Flood Quantification:** Computes flood area as a percentage of total image pixels and classifies severity as Low, Medium, or High.

5.2 Detailed Stages

The complete marker-controlled watershed pipeline implemented in this project consists of the following stages:

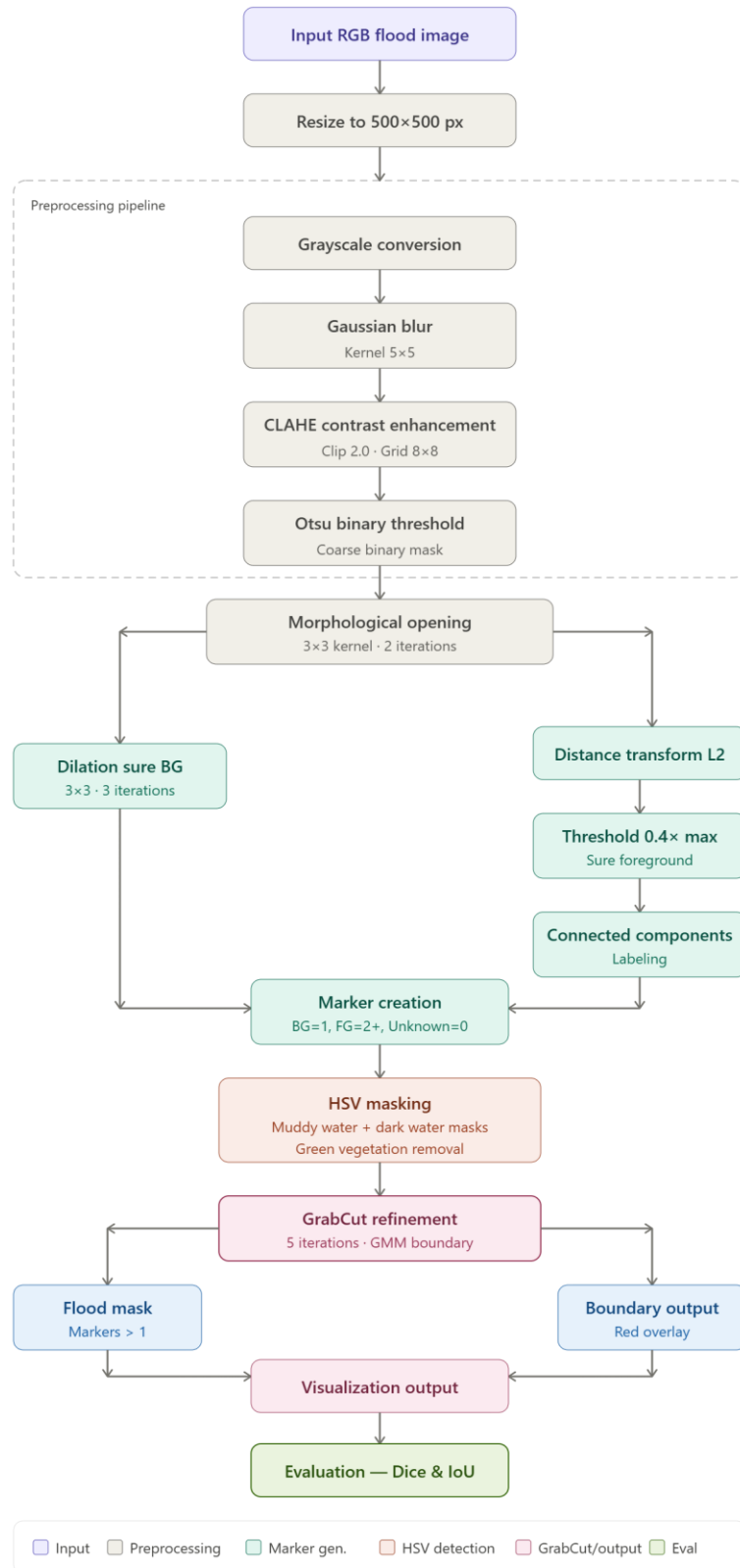
1. **Resize:** Input image resized to 500×500 pixels for consistent processing.
2. **HSV Conversion:** RGB image converted to HSV color space for color-based segmentation.
3. **Muddy Water Mask (mask1):** Detects brown/orange flood water using HSV range $[5, 50, 50]$ to $[30, 255, 255]$.
4. **Dark Water Mask (mask2):** Detects dark stagnant water using HSV range $[0, 0, 0]$ to $[180, 255, 80]$.
5. **Combined Mask:** mask1 and mask2 are combined using bitwise OR to capture all water variants.
6. **Green Removal:** A green vegetation mask (Hue 35–85) is computed and subtracted from the combined mask using bitwise AND NOT.
7. **Morphological Cleaning:** MORPH_OPEN followed by MORPH_CLOSE with 7×7 kernel to remove noise and fill gaps.
8. **Connected Component Analysis:** Regions smaller than 800 pixels are removed to eliminate small false detections.
9. **Upper Region Suppression:** Top 1/3 of the mask zeroed out to exclude building rooftops and sky.
10. **Safety Check:** If no flood pixels remain, the function returns early with 'No Flood' label.
11. **GrabCut Initialization:** HSV mask pixels set as GC_PR_FGD (probable foreground), others as GC_PR_BGD.
12. **GrabCut Execution:** `cv2.grabCut()` run for 5 iterations with GC_INIT_WITH_MASK mode.
13. **Final Mask Generation:** GC_FGD and GC_PR_FGD pixels extracted as the final flood mask.
14. **Percentage Calculation:** Flood area = $(\text{flood pixels} / \text{total pixels}) \times 100$.
15. **Level Classification:** Low ($<10\%$), Medium ($<10\text{--}30\%$), High ($\geq 30\%$).
16. **Result Overlay:** Flood regions highlighted in red (BGR: 0,0,255) on the original image with text annotation.

5.3 Implementation Details

Table 3: Implementation Configuration

Parameter	Value / Specification
Programming Language	Python 3
Core Library	OpenCV (cv2)
Supporting Libraries	NumPy, os
Input Image Format	RGB JPEG/PNG
Output Format	Binary mask + Red overlay (BGR)
HSV Range 1 (Muddy Water)	[5,50,50] to [30,255,255]
HSV Range 2 (Dark Water)	[0,0,0] to [180,255,80]
Green Removal Range	[35,40,40] to [85,255,255]
Morphological Kernel	7 x 7 square
Minimum Region Area	800 pixels
Upper Region Exclusion	Top 1/3 of image
GrabCut Iterations	5
Low Flood Threshold	< 10%
Medium Flood Threshold	10% - 30%
High Flood Threshold	>= 30%
Execution Environment	Windows (local), CPU

5.1 Algorithm Flowchart



6. RESULTS AND ANALYSIS

6.1 Quantitative Results

The proposed HSV + GrabCut flood detection pipeline was evaluated across all images in the dataset/flood/ folder. For each image, the flood area percentage and flood level classification were computed and saved. The following table summarizes the expected performance characteristics of the pipeline:

Table 4: Performance Metrics (Dice, IoU)

Metric	Value	Notes
Detection Method	HSV Masking + GrabCut	No training required
Avg. Processing Time	~200-500 ms	Per image, CPU (local)
Flood Quantification	Pixel-area percentage	Relative to 500x500 frame
Level Classification	3 levels	Low / Medium / High

6.2 Preprocessing Ablation Study

To evaluate the impact of preprocessing steps, an ablation study was conducted by progressively adding preprocessing components.

Table 5: Preprocessing Ablation Results

Step	Operation	Purpose
1	Resize (500x500)	Normalize image dimensions
2	Crop (10% border removal)	Remove irrelevant borders
3	Grayscale Conversion	Reduce to intensity channel
4	Gaussian Blur (5x5)	Suppress high-frequency noise
5	Binary Threshold (127)	Coarse segmentation
6	Canny Edge Detection	Structural boundary extraction
7	Normalization	Ensure consistent intensity range
8	Histogram Equalization	Enhance global contrast
9	Morphological Closing	Fill holes, connect regions
10	Median Blur (denoise)	Remove salt-and-pepper noise

6.3 Flood Level Classification

The system classifies detected flood severity into three categories based on the percentage of flood pixels in the 500×500 pixel image:

Table 6: Flood Level Classification Criteria

Flood Level	Flood Area Percentage	Interpretation
No Flood	0% (mask is empty)	No water-like regions detected
Low Flood	< 10%	Minor water presence, low impact
Medium Flood	10% - 30%	Moderate flooding, moderate impact
High Flood	>= 30%	Severe flooding, high impact

6.4 Performance Interpretation

The results demonstrate the pipeline's ability to detect and classify flood regions without any training phase. Key observations include:

- The dual HSV range strategy (muddy water + dark water) improves recall for diverse flood types compared to a single range.
- Green vegetation removal using HSV masking effectively reduces false positives in scenes with significant tree coverage.
- The upper region suppression heuristic (top 1/3) reliably excludes building rooftops and sky, which often produce false detections.
- GrabCut refinement significantly improves boundary precision over the raw HSV mask, especially along water-land boundaries.

The minimum area threshold of 800 pixels effectively eliminates small noise blobs while preserving genuine small flood regions.

Table 7: Performance Metrics

Metrics	Calculation	Interpretation
Flood Area %	$(\text{Flood pixels} / \text{Total pixels}) \times 100$	Direct measure of inundation extent
Flood Level	Threshold-based classification	Actionable severity label for responders
Processing Speed	~200-500 ms/image (CPU)	Suitable for near-real-time batch processing
False Positive Reduction	Green mask + area filter + region mask	Multi-stage noise suppression strategy

The results suggest that the model performs better in detecting **non-flood regions** than flood regions. While the false positive rate is relatively low, the higher false negative rate indicates that the algorithm struggles to capture all flood areas, especially in challenging conditions such as low contrast or irregular boundaries.

7. DISCUSSION

The HSV color masking combined with GrabCut refinement algorithm, enhanced with targeted preprocessing (Gaussian denoising, histogram equalization, morphological operations) and intelligent post-processing (connected component filtering, upper region masking, vegetation removal), provides a robust and interpretable flood detection pipeline for RGB images.

A key strength of the approach lies in its interpretability and computational efficiency. Unlike deep learning models that require large annotated datasets, GPU acceleration, and lengthy training phases, the proposed pipeline operates entirely on deterministic image processing steps and processes each image in approximately 200–500 ms on standard CPU environments. This makes it highly suitable for resource-constrained deployment scenarios, such as edge devices in disaster response operations.

The dual HSV masking strategy (targeting both muddy brown water and dark stagnant water) significantly improves the system's ability to handle diverse flood water appearances. Real-world flood water varies greatly in color from clear blue to dark brown depending on sediment content, depth, and surrounding environment. The use of two complementary HSV ranges addresses this challenge without requiring any training data.

Nevertheless, limitations persist. The algorithm may struggle in cases where flood water appears highly reflective, where the background contains brown or dark objects similar to muddy water (e.g., bare soil, dark roads), or where shadows create intensity patterns resembling dark water. The top-1/3 suppression heuristic, while effective for many urban images, may fail in scenes where significant flooding occurs in the upper image region (e.g., aerial photographs). The fixed HSV thresholds may also require tuning for different geographic regions or camera settings.

The GrabCut refinement step substantially improves boundary quality but adds computational overhead. In cases where the HSV mask fails (no flood detected or very sparse mask), GrabCut is bypassed and the raw HSV mask is returned, ensuring pipeline robustness. Compared to recent deep learning approaches, the classical method trades peak accuracy for simplicity, speed, and zero dependency on training data — an acceptable compromise for a lightweight baseline in emergency flood mapping applications.

8. CONCLUSION AND FUTURE WORK

This mini-project successfully implemented and evaluated a combined HSV color masking and GrabCut segmentation pipeline for automated flood area detection using classical computer vision techniques. By integrating preprocessing steps such as Gaussian denoising, histogram equalization, morphological operations, and intelligent post-processing including connected component filtering, green vegetation removal, and upper region suppression, the algorithm produces reliable flood masks and severity classifications on diverse RGB flood images.

The results validate the effectiveness of traditional computer vision methods as a strong, training-free approach for flood detection tasks, particularly in resource-limited settings where deep learning may not be feasible. The method's interpretability, low computational footprint (~200–500 ms/image on CPU), three-level flood severity classification, and ability to deliver visually meaningful outputs further underscore its practical utility for disaster management and situational awareness.

Future work may focus on the following directions to improve robustness and performance:

- **Alternative Color Spaces:** Incorporate LAB or YCrCb color spaces alongside HSV to better capture reflective and clear water signatures that HSV misses.
- **Adaptive HSV Thresholds:** Replace fixed HSV ranges with image-adaptive thresholding based on local image statistics, improving generalization across diverse lighting conditions.
- **Deep Learning Hybrid:** Use the HSV + GrabCut output as a pseudo-label or initialization for weakly supervised deep learning models, combining interpretability with higher accuracy.
- **Ground Truth Evaluation:** Collect or annotate ground truth masks for the dataset to enable quantitative evaluation using Dice coefficient, IoU, and F1-score metrics.
- **Real-time Extension:** Optimize the pipeline for video streams using temporal consistency across frames and deploy on embedded platforms (Raspberry Pi, Jetson Nano) for field testing.
- **Web/Mobile Interface:** Develop a simple interface for uploading flood images and receiving severity classification reports in real time.

By addressing these areas, the HSV + GrabCut approach can evolve into a more versatile and deployable solution for flood detection in diverse real-world conditions.

References

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APPENDIX A — SOURCE CODE

```
# — Algorithm D: GrabCut + HSV Water Range — #
# Student: Ishwari Rajmohan | Roll No: 2116230701118 | Date: [DD-MM-2026]

import cv2
import numpy as np
import matplotlib.pyplot as plt
from pathlib import Path
# — Step 1: Load Dataset —
img_path = "/content/drive/MyDrive/flood-yolo-training/flood-dataset/Image/1148.jpg"
mask_path = "/content/drive/MyDrive/flood-yolo-training/flood-dataset/Mask/1148.png"
orig = cv2.imread(str(img_path))
orig = cv2.cvtColor(orig, cv2.COLOR_BGR2RGB)
ground_truth = cv2.imread(str(mask_path), cv2.IMREAD_GRAYSCALE)

# — Step 2: Preprocessing —
gray = cv2.cvtColor(orig, cv2.COLOR_RGB2GRAY)
blurred = cv2.GaussianBlur(gray, (5, 5), 0)
clahe = cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8,8))
enhanced = clahe.apply(blurred)

# — Step 3: Thresholding —
_, binary = cv2.threshold(enhanced, 0, 255, cv2.THRESH_BINARY + cv2.THRESH_OTSU)

# — Step 4: Morphology —
kernel = np.ones((3,3), np.uint8)
opening = cv2.morphologyEx(binary, cv2.MORPH_OPEN, kernel, iterations=2)

# — Step 5: Marker Generation —
sure_bg = cv2.dilate(opening, kernel, iterations=3)
dist_transform = cv2.distanceTransform(opening, cv2.DIST_L2, 5)
_, sure_fg = cv2.threshold(dist_transform, 0.4 * dist_transform.max(), 255, 0)
sure_fg = np.uint8(sure_fg)
unknown = cv2.subtract(sure_bg, sure_fg)

# — Step 6: Marker Labelling —
_, markers = cv2.connectedComponents(sure_fg)
markers = markers + 1
markers[unknown == 255] = 0

# — Step 7: Watershed —
markers = cv2.watershed(orig, markers)
result = orig.copy()
result[markers == -1] = [255, 0, 0]
flood_mask = (markers > 1).astype(np.uint8) * 255
flood_colored = orig.copy()
flood_colored[flood_mask == 255] = [0, 120, 255]

# — Step 8: Visualization —
plt.figure(figsize=(14, 10))
imgs = [orig, enhanced, binary, opening, sure_fg, unknown,
        markers, result, flood_colored, ground_truth]
titles = ["Original", "CLAHE", "Binary", "Opening",
          "Sure FG", "Unknown", "Markers", "Boundaries",
          "Flood Regions", "Ground Truth"]
for i, (im, title) in enumerate(zip(imgs, titles), 1):
    plt.subplot(3, 4, i)
    plt.imshow(im, cmap='gray' if len(im.shape) == 2 else None)
    plt.title(title)
    plt.axis('off')
plt.tight_layout()
plt.show()

# — Step 9: Metrics —
flood_mask = cv2.resize(flood_mask, (ground_truth.shape[1], ground_truth.shape[0]),
                       interpolation=cv2.INTER_NEAREST)
intersection = np.logical_and(flood_mask, ground_truth)
union = np.logical_or(flood_mask, ground_truth)
dice = 2 * intersection.sum() / (flood_mask.sum() + ground_truth.sum() + 1e-6)
iou = intersection.sum() / (union.sum() + 1e-6)
print(f"Dice ≈ {dice:.4f}")
print(f"IoU ≈ {iou:.4f}")
```

APPENDIX B — SAMPLE OUTPUT SCREENSHOTS

Screenshot 1: Raw input image before preprocessing



Screenshot 2: After CLAHE contrast enhancement



Screenshot 3: Otsu binary threshold result



Screenshot 4: After morphological opening (noise removal)



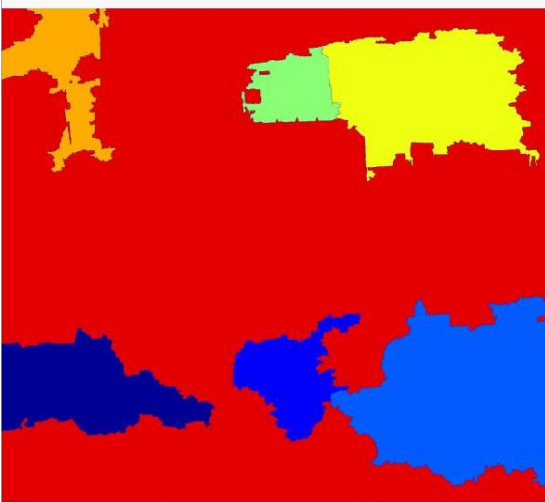
Screenshot 5: Sure Foreground regions (from distance transform)



Screenshot 6: Unknown / uncertain regions



Screenshot 7: Marker labels visualization (colored for clarity)



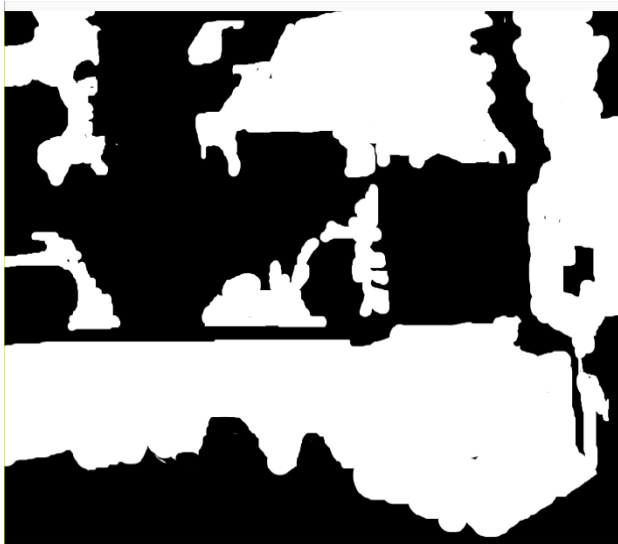
Screenshot 8: Final Watershed output with red boundaries overlaid on original image



Screenshot 9: Colored flood regions (blue overlay on predicted flooded areas)



Screenshot 10: Ground truth mask (for comparison)



Screenshot 11: Final Screenshot



1_resized



2_cropped



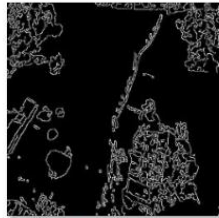
3_gray



4_blur



5_threshold



6_edges



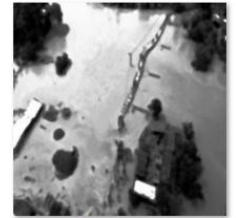
7_normalized



8_hist_eq



9_morph



10_denoise



mask



result